

16.1 Mathematical Foundation

16.1.1 Domain

$\Omega \subset \mathbb{R}^d$, Ω is compact

i.e., Ω is closed and bounded, containing all admissible system states.

$\Psi \in \Omega \Rightarrow \|\Psi\| \leq M$

for some finite constant $M > 0$.

16.1.2 State Variable

$\Psi_k \in \Omega$

The state variable is assumed to be:

- measurable
- bounded

16.1.3 Potential Function

$V \in C^2(\Omega), \quad \nabla^2 V(x) \succeq m I, \quad m > 0$

Thus, $V(x)$ is strongly convex, implying:

- existence of a unique global minimum
- absence of multiple equilibria

16.1.4 Energy Functional

$$\mathcal{E}(\Psi) = V(\Psi) + \frac{1}{2} \|\nabla \Psi\|^2$$

which is a convex functional.

16.1.5 Probability Space

$$P(x,t) \geq \varepsilon > 0, \quad \int_{\Omega} P(x,t) dx = 1$$

ensuring:

- absence of singularities
- well-defined logarithmic terms $\ln P$

16.1.6 Lipschitz Continuity

$$\|F(x) - F(y)\| \leq L \|x - y\|$$

for some $L > 0$, ensuring regularity of the system dynamics.

16.1.7 Existence and Uniqueness

Under the above assumptions, the system admits a unique solution trajectory:

by the Picard–Lindelöf theorem.

16.1.8 Time Discretization

$$t_k = k\,dt, \quad dt < \frac{1}{L}$$

ensuring numerical stability of the discrete-time approximation.

Closure Statement (Part I)

$$\boxed{\text{The system is well-posed, bounded, and admits a unique solution.}}$$

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